SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club | Submission deadline: 25 October 2026

Instructions: Complete every field. Submit as a single PDF named **SDC2026_TeamName_Preliminary.pdf** before the deadline. Answers must comply with the word limits. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Field	Response
Team name	Empathy Coders
Selected domain	Domain 2 - Orbyte Smart SME Challenge
Selected SDG track (1-17)	SDG 8 - Decent Work and Economic Growth (sub-targets 8.2 and 8.3)
Project / solution name	OrderPilot - Smart Order Consolidation and Approval for SMEs
Name (leader)	Yang Jiahui
Email & phone (leader)	[TO BE COMPLETED BEFORE SUBMISSION]
University affiliation(s)	Xiamen University Malaysia (XMUM)
Number of team members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context [maximum 250 words]

Malaysian micro and small businesses rarely receive orders through a tidy form. A cafe supplier gets a WhatsApp message for 20 boxes of coffee beans, an email for 40 sleeves of cups, a follow-up that adds 10 more boxes, and a correction that changes the quantity. Ads, price-list requests and supplier pitches arrive in the same inbox. The owner reads everything, copies it into Excel, checks stock, calculates cost and margin, and guesses whether a supplier can restock before the promised delivery date. This work happens after closing, on a phone, with no audit trail. The burden is not only time. A missed message becomes a missed sale. A copied quantity becomes a wrong delivery. A quote below cost is discovered after the invoice. A stock shortage is discovered after the customer has been promised a date. Staff with different levels of digital confidence perform the same task differently, and there is no reliable record of who approved what. The target users are owners and small teams of Malaysian micro and small businesses in food service, retail and distribution. They already use WhatsApp, email and spreadsheets; they cannot afford a custom ERP or weeks of training. They need one reliable, easy-to-learn workflow that connects the order to the business data, surfaces the risk, keeps a human in control and leaves a record. That is the gap OrderPilot addresses.

2.2 The Proposed AI Solution [maximum 300 words]

OrderPilot is a web application that turns WhatsApp, email and manual messages into one verified order desk. A message enters the inbox, and AI classifies it as an order, an order update, an inquiry, an advertisement, a supplier message, a payment reminder or spam. For orders, AI extracts the customer, contact, items, quantities, unit prices, delivery date, address and notes. Follow-up messages are consolidated into the same order: add, replace, remove or update the delivery date. AI is used only where it adds value: reading unstructured language. Stock, cost, margin, delivery feasibility and risk are calculated by deterministic rules. The rules compare required quantity with available stock (on hand minus reserved), flag safety-stock warnings, check supplier lead time against the delivery date, calculate revenue, cost, gross profit and margin, detect a quote below cost or a price deviation, find possible duplicates, and list missing fields. The result is a transparent recommendation: Ready to accept, Procurement needed, Review recommended, Customer info needed, or High risk. The owner or manager sees one table, not a message-by-message queue. Each row is an order with the customer, items, amount, delivery, stock, flags and recommendation. The approval is on the far right. The server enforces roles: staff can review and request information, but only the owner/manager can approve, reject or edit. High-risk approvals require a written justification. Approval reserves stock, creates internal purchase drafts for shortages, and writes an order record and an activity entry. Every

import, analysis, approval, rejection and stock change is logged and can be exported to CSV or Excel. The solution applies the SDC workshops: a GenAI API for extraction, a repository/data layer for business records, and Replit hosting and deployment. A server-side adapter isolates the AI provider, with a deterministic fallback if the API is unavailable.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

The prototype is a functional, deployed web application. The stack is deliberately small so the team can explain and operate every layer.

Layer	Technology	Purpose
Frontend	HTML5, CSS3, vanilla JavaScript (ES modules)	Order Desk, Inbox, Orders, Activity, Catalog, Settings
Backend	Node.js 20+, Express 4	REST API, RBAC, triage orchestration, exports
Data	JSON files (seed committed; runtime gitignored)	Customers, products, stock, suppliers, messages, orders, drafts, activity
AI	DeepSeek chat/completions API (OpenAI-compatible)	Classification and field extraction only
Fallback	Deterministic rules extractor	Keeps the demo working without an API key
Exports	ExcelJS + CSV writer	Order and activity exports
Hosting	Replit (Core credit)	Required deployment platform
Repository	GitHub, Apache License 2.0	Public source and commit history
GPU	None	No training, fine-tuning or GPU inference; RunPod not used

System block diagram (text version)

Layer	Flow	Notes
1. Input	WhatsApp message -> Email -> Manual paste or upload	Synthetic messages only; no real messaging API
2. Interface	Browser UI (Order Desk / Inbox / Orders / Activity) -> REST API	One tidy order table; no message-by-message approval
3. Triage	Express triage service -> AI adapter (DeepSeek) -> JSON extraction	Rules fallback produces the same shape if AI fails
4. Decision	Order/approval service -> pure decision engine	Stock, shortage, lead time, revenue, cost, margin, price, duplicates, missing fields
5. Action	Human approval -> stock reservation -> purchase drafts -> activity log	Owner/Manager only; high-risk approvals need a note
6. Data	JSON repositories (seed committed, runtime gitignored) <- OrbyteAdapter (mock)	Replaceable by SQLite/Postgres or a real Orbyte Connect adapter

Official SDC sponsor-provided GenAI API or Cloud Platform

We used the **Replit Core credit** for development and deployment. We did **not** use RunPod because the project has no GPU workload. We did not receive Orbyte production access; the Orbyte integration is a mock adapter over synthetic data, as required by the brief. The GenAI API is a team-selected external provider: **DeepSeek** (chat/completions). The exact provider and model are declared in the Technical & Tools Declaration.

3.2 Design-Thinking & Deployment Strategy [maximum 200 words]

We framed the problem around one recurring moment: the owner opening WhatsApp after closing and rebuilding the day's orders in Excel. We mapped that journey - message, interpretation, data lookup, stock check, margin calculation, delivery promise, approval, record - and removed every step that did not need a human. The first prototype was a

message list with extracted fields; testing it showed that the owner still had to compare messages one by one. We changed the primary object from message to consolidated order and made one table the default screen. We then tested the rules against a synthetic 29-message set covering clear orders, Malay/Manglish, add-ons, corrections, ads, inquiries, supplier pitches, payment reminders and spam. The design is deliberately easy to learn: plain-language labels, visible reasons, no configuration before the first order, and a single approval action on the right. Deployment is Replit-first: the app has no build step, starts with npm start, seeds its own synthetic data, and provides a manager-only demo reset. A deterministic extraction fallback protects the live demo from API or network failure.

4. Preliminary Environmental & SDG Impact Framework

The target, baseline and KPI below are used by judges to assess Innovation and Creativity as well as Feasibility of build. The baseline is an internal synthetic benchmark, not a published statistic; the measurement method is reproducible from the repository.

Metric category	Your response
Target SDG sub-target(s)	**SDG 8.2** - achieve higher levels of productivity through technological upgrading and innovation. **SDG 8.3** - promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalisation and growth of micro-, small- and medium-sized enterprises.
Current problem baseline	In our synthetic 12-message benchmark, manual reading, copying into a spreadsheet, stock checking and margin calculation took approximately **18 minutes per batch**. About **2 of 12** messages required a follow-up because a required field was missing, and **5 of 12** were non-orders that still consumed attention. This is an internal synthetic baseline, not a published statistic.
Core measurable KPI	(1) Minutes to process a 12-message batch. (2) Extraction/field error rate against a manually verified ground truth. (3) Number of missed orders in the test set. (4) Percentage of approvals with a complete actor + timestamp + reason audit entry. (5) Percentage of orders where stock and margin are checked before approval.
Phase 2 impact target	**60% reduction** in manual order-processing time per 12-message batch; **<2%** extraction error rate on the verified test set; **0 missed orders**; **100%** of approvals recorded with actor, timestamp and audit trail; **100%** of approved orders pre-checked for stock and margin.
Measurement method	Replay the same synthetic message set through the manual process and OrderPilot; compare elapsed time, field-level accuracy and missed orders. Record the rules and audit trail for every decision. The synthetic dataset and test script are in the repository (data/seed/ and tests/).

Why this matters for SDG 8

A small business that spends less time copying orders can spend more time serving customers and growing. Fewer stockouts and wrong quotes protect the margin that keeps a micro-business viable. The workflow is designed for staff with different digital confidence, so adoption does not depend on formal training. The Orbyte adapter approach allows the same capability to reach many SMEs without a custom build. A Phase 2 environmental opportunity is better stock visibility reducing over-ordering and waste, especially for perishable cafe supplies; this is not claimed as a quantified Phase 1 impact.

5. Eligibility & Readiness Checklist

[ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

[x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

[x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

[x] All fields in this template have been completed within the word limits stated.

Team note: tick the workshop box after every member has confirmed attendance. The prototype is deployed on Replit before submission.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

[ ] Yes

[x] No

If yes, describe: No pre-existing code, design, research or other work created before the hackathon is included in the core solution. The Apache License 2.0 LICENSE file is the standard license text; the official submission template and sample repository structure were provided by the organisers.

6.2 Third-Party Code, Datasets, or Models

[x] Yes

[ ] No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type	License	Use
Node.js 20+	Code / runtime	MIT	Server runtime
Express 4.x	Code / framework	MIT	REST API
ExcelJS 4.x	Code / library	MIT	XLSX export
dotenv 16.x	Code / library	BSD-2-Clause	Local environment loading
DM Sans, Fraunces	Font asset	SIL OFL 1.1	UI typography (Google Fonts)
DeepSeek deepseek-flash	Model / API	Provider terms	Message classification and field extraction only
Synthetic SME dataset	Dataset	Team-created	Demo customers, products, stock, suppliers, messages

No public or private dataset was used for training or fine-tuning. No real personal or company data is included. The AI model is pre-trained and accessed through an API; it is not fine-tuned or self-trained by the team.

6.3 Employment-Related IP Conflicts

[ ] Yes

[x] No

If yes, describe: None declared. The team confirms that no member is bound by an employment, internship or consulting agreement that would give an employer ownership of this hackathon work.

Team confirmation required: every member must confirm the IP and employment declarations before the leader signs.

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: _____ **Date:** 21 October 2026

Name: Yang Jiahui (Team Leader) **Team:** Empathy Coders

Signature: the team leader signs here before submitting. Date: 21 October 2026.